

Fraud Velocity Detection Policy

SentinellQ Internal Policy Document - Confidential

1. Purpose and Scope

This policy defines the detection rules, thresholds, and investigation procedures for velocity-based fraud patterns across all four partner banks (HDFC, SBI, ICICI, Axis). Velocity fraud occurs when a threat actor executes multiple transactions in rapid succession from a single compromised account, often to drain funds before detection systems can respond. This document applies to all real-time and batch transaction monitoring pipelines operated by the SentinellQ platform.

2. Definitions

Velocity Burst: Three or more transactions originating from the same `account_id` within any rolling 60-minute window. **Micro-Burst:** Five or more transactions within a 15-minute window, considered a high-severity sub-pattern. **Rolling Window:** A sliding time frame anchored to the most recent transaction timestamp for a given account. **Cooling Period:** A mandatory 24-hour observation window imposed on an account after a velocity burst is confirmed.

3. Detection Rules

Rule V-1: If an account generates 3 or more outgoing transactions within a 60-minute rolling window, the `velocity_flag` SHALL be set to True for all transactions in that window. The system counts the current transaction plus all prior transactions from the same account whose timestamps fall within `[current_timestamp - 60 minutes, current_timestamp]`. **Rule V-2:** If an account generates 5 or more transactions within a 15-minute window, the alert severity SHALL be escalated to CRITICAL regardless of individual transaction amounts. **Rule V-3:** Velocity bursts originating during unusual hours (00:00-04:59) receive an additional risk multiplier of 1.3x applied to the base score. **Rule V-4:** Velocity bursts involving round amounts (multiples of 500 INR, minimum 10,000 INR) compound with the `round_amount_flag` and increase the combined risk score.

4. Thresholds and Scoring Impact

A confirmed velocity burst adds a base contribution of 20 points to the transaction risk score. Combined with other active flags, this contribution is additive. The scoring formula is: $\text{base_score} = \text{active_flag_count} * 20$. When the `velocity_flag` is combined with `unusual_hour_flag`, the night multiplier of 1.3x is applied after the base calculation. Threshold boundaries:

- Score ≥ 85 : CRITICAL tier -- immediate account hold and Gmail alert
- Score 70-84: HIGH tier -- added to analyst approval queue
- Score 40-69: MED tier -- queued for daily batch review at 18:00
- Score < 40 : LOW tier -- auto-cleared

with pattern hash stored. Confidence is assessed as HIGH when the LLM provides supporting policy evidence, and LOW when rule-based fallback scoring is used.

5. Investigation Procedures

Step 1: Upon `velocity_flag` activation, the system retrieves the full transaction history for the affected `account_id` from the current dataset. **Step 2:** The investigating analyst reviews the temporal clustering of transactions, noting the time gaps between consecutive transactions. **Step 3:** Cross-reference `merchant_ids` involved in the burst -- if the same merchant appears in multiple burst transactions, flag as potential merchant collusion pattern. **Step 4:** Verify geographic consistency -- if velocity burst transactions span multiple distant locations, escalate to combined velocity+geo pattern. **Step 5:** For CRITICAL tier velocity cases, the system auto-generates an account hold recommendation and dispatches a Gmail alert to the configured notification address within 2 minutes of detection. **Step 6:** Document findings in the investigation report (Section 3: CRITICAL Alerts or Section 4: HIGH-Risk Transactions as appropriate).

6. Relay Chain and Circular Pattern Detection

Velocity bursts are often precursors to relay chain fraud, where funds move rapidly through a sequence of intermediary accounts. When the Transaction Agent builds the network graph, accounts exhibiting velocity bursts are given priority in the relay detection algorithm. A relay node is defined as any account or merchant node with combined in-degree plus out-degree exceeding 5 in the directed transaction graph. Circular chains are detected via NetworkX `simple_cycles`

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algorithm. Accounts participating in circular chains with velocity bursts receive the maximum compound risk score. The network graph visualization on Page 2 highlights relay nodes in red and cycle-participating nodes with a pulsing animation.

7. Pattern Memory Integration

When an analyst approves or rejects a velocity-flagged transaction via the approval panel, the decision is recorded in the Pattern Store. The pattern hash is computed from the sorted active flag names, bank name, and hour bucket ($\text{floor}(\text{hour}/6)$). Approved patterns receive a +10 score adjustment on future occurrences. Rejected patterns receive a -15 score adjustment. Memory decay applies: full adjustment within 90 days, 50% from 91-180 days, and 25% beyond 180 days. This ensures the system adapts to evolving fraud patterns while not over-weighting stale decisions.

8. Compliance and Audit Requirements

All velocity detections must be logged with timestamp, account_id, transaction count in window, and resulting tier assignment. The PDF investigation report must include velocity pattern findings in the appropriate section. Reports are uploaded to S3 bucket fraud-investigation-group9 under the path reports/{date}/{session_id}/. Quarterly review of velocity thresholds is mandatory. The current threshold of 3 transactions in 60 minutes was calibrated against historical fraud data from all four partner banks and may be adjusted based on false positive rates.